

Understanding Topics in News Coverage of Video-Based Social Media and Youth Mental Health

Multi-Label Topic Classification

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Abstract

The proliferation of video-based social media platforms such as YouTube, TikTok, Snapchat, and Instagram Reels has raised significant concerns about their impact on youth mental health. Understanding how news media covers this intersection is crucial for policymakers, clinicians, and researchers. However, manually analyzing large volumes of news articles to identify relevant mental health topics is time-consuming and subjective. In this work, we present an automated pipeline for multi-label topic classification of news articles discussing video-based social media and youth mental health. Starting from 1,998 raw articles collected from news sources, we employ a zero-shot learning approach using BART-MNLI for relevance filtering and multi-label topic annotation across 33 predefined mental health topics. We compare three classification approaches: a classical TF-IDF with Logistic Regression baseline, a standard DistilBERT transformer model, and an improved transformer with class weighting and optimized decision thresholds. Our improved transformer model achieves a macro F1-score of 0.507, successfully predicting 31 out of 33 topics compared to only 8 topics in the naive transformer approach. We identify key topic clusters including social media usage (91.0% of articles), resilience/coping mechanisms (84.6%), and screen time effects (82.1%). Our findings reveal that news coverage predominantly focuses on prevalent topics while underrepresenting critical areas such as foster care, sleep issues, and eating disorders. This work demonstrates the feasibility of automated topic extraction for mental health discourse analysis and provides insights into media representation patterns that can inform public health communication strategies.

1 Introduction

The widespread adoption of video-based social media platforms such as YouTube, TikTok, Snapchat, and Instagram Reels among adolescents has raised significant concerns about their impact on youth mental health. News media coverage of this issue is extensive but diverse,

spanning topics from anxiety and depression to school policies and access to mental health care. Manually analyzing and categorizing these articles to understand coverage patterns is time-consuming and impractical for large datasets.

This project develops an automated pipeline for multi-label topic classification of news articles discussing video-based social media and youth mental health. Starting with 1,998 raw articles collected from news sources, we apply a series of preprocessing steps including deduplication, relevance filtering, and automated topic annotation. The final dataset consists of 864 articles annotated across 33 predefined mental health topics such as anxiety, depression, social media usage, bullying, and access to care.

Our primary objective is to compare the performance of different machine learning approaches for this multi-label classification task. We implement and evaluate three models: a classical TF-IDF with Logistic Regression baseline, a standard DistilBERT transformer model, and an improved transformer incorporating techniques to handle class imbalance. The key challenge is that topic frequencies vary dramatically, with some topics appearing in over 90

We utilize zero-shot learning with the BART-MNLI [1] model for automated annotation, eliminating the need for manual labeling. Our experiments demonstrate that while the TF-IDF baseline achieves strong performance with a macro F1-score of 0.591, a naive transformer implementation severely underfits minority classes, predicting only 8 of 33 topics. By applying class weighting and per-topic threshold optimization, we improve the transformer to predict 31 of 33 topics with a macro F1-score of 0.507. Additionally, we analyze topic distribution patterns and visualize article clusters to understand how news media frames the relationship between video-based social media and youth mental health.

2 Problem Formulation

2.1 Task Definition

Our task is multi-label text classification, where each news article can be assigned multiple topic labels simultaneously. Unlike traditional multi-class classification where each sample belongs to exactly one class, multi-label classification allows an article to be relevant to several mental health topics at once.

Formally, let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$ denote our dataset of N articles, where:

- x_i represents the i -th article as a text document
- $y_i \in \{0, 1\}^L$ is a binary label vector of length $L = 33$
- $y_i^{(j)} = 1$ if article x_i discusses topic j , otherwise $y_i^{(j)} = 0$

The set of 33 topic labels $\mathcal{L} = \{l_1, l_2, \dots, l_{33}\}$ includes mental health topics such as `anxiety`, `depression`, `social media`, `bullying/cyberbullying`, `screen time`, `access to care`, and others.

2.2 Model Objective

The goal is to learn a classification function $f : \mathcal{X} \rightarrow [0, 1]^L$ that takes an article x as input and outputs a probability vector $\hat{y} = f(x)$, where $\hat{y}^{(j)}$ represents the predicted probability that article x is relevant to topic l_j .

To convert probabilities into binary predictions, we apply a threshold function:

$$\hat{y}_{pred}^{(j)} = \begin{cases} 1 & \text{if } \hat{y}^{(j)} \geq \tau_j \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

where τ_j is the decision threshold for topic j . Standard practice uses $\tau_j = 0.5$ for all topics. However, we also experiment with optimizing individual thresholds $\{\tau_j\}_{j=1}^L$ on a validation set to improve performance on imbalanced classes.

2.3 Evaluation Metrics

We evaluate our models using four standard multi-label classification metrics:

Exact Match Ratio measures the percentage of articles where all labels are predicted correctly:

$$ExactMatch = \frac{1}{N} \sum_{i=1}^N 1[y_i = \hat{y}_{pred,i}] \quad (2)$$

Hamming Loss calculates the fraction of incorrect label predictions:

$$HammingLoss = \frac{1}{NL} \sum_{i=1}^N \sum_{j=1}^L 1[y_i^{(j)} \neq \hat{y}_{pred,i}^{(j)}] \quad (3)$$

Micro-averaged F1 computes precision and recall globally across all labels and samples:

$$F1_{micro} = \frac{2 \cdot P_{micro} \cdot R_{micro}}{P_{micro} + R_{micro}} \quad (4)$$

where P_{micro} and R_{micro} are calculated by aggregating true positives, false positives, and false negatives across all topics.

Macro-averaged F1 computes F1-score for each topic independently and averages them:

$$F1_{macro} = \frac{1}{L} \sum_{j=1}^L F1^{(j)} \quad (5)$$

Macro F1 is particularly important for our task because it treats all topics equally regardless of frequency. This metric reveals whether models can predict rare topics (e.g., foster care, eating disorders) in addition to common ones (e.g., social media, screen time).

2.4 Class Imbalance Problem

A key challenge in our dataset is extreme class imbalance. Let $n_j = \sum_{i=1}^N y_i^{(j)}$ represent the number of articles labeled with topic j . In our data, frequencies range from $n_{social_media} = 786$

(91.0% of articles) to $n_{foster_care} = 9$ (1.0% of articles). This two order of magnitude difference causes models trained with standard loss functions to focus on frequent topics while ignoring rare ones.

To address this, we use weighted Binary Cross-Entropy loss:

$$\mathcal{L}_{BCE}(y, \hat{y}) = -\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^L w_j \left[y_i^{(j)} \log(\hat{y}_i^{(j)}) + (1 - y_i^{(j)}) \log(1 - \hat{y}_i^{(j)}) \right] \quad (6)$$

where w_j is the positive class weight for topic j , computed using inverse frequency:

$$w_j = \min \left(10.0, \max \left(0.5, \frac{N}{2(n_j + 1)} \right) \right) \quad (7)$$

This formula assigns higher weights to rare topics, forcing the model to pay more attention to them during training. We cap weights at 10.0 to prevent training instability.

3 Methods

3.1 Pipeline Overview

Our approach consists of four stages: (1) data collection and preprocessing, (2) relevance filtering, (3) automated topic annotation, and (4) classification model training. This pipeline processes raw news articles into a labeled dataset suitable for multi-label classification.

3.2 Data Collection and Preprocessing

We collected 1,998 raw news articles from YouTube-related (1,326 articles) and Snapchat-related (676 articles) news sources. The preprocessing pipeline performs text normalization, platform detection (scanning for mentions of YouTube, TikTok, Snapchat, Instagram Reels, Shorts), deduplication using MD5 hashing of title and body text, and filtering articles with fewer than 30 words. Manual quality control removed irrelevant content, advertisements, and formatting errors, reducing the dataset to 870 high-quality articles.

3.3 Zero-Shot Relevance Filtering

We use BART-MNLI (`facebook/bart-large-mnli`) for zero-shot classification to filter articles where both video-based social media and youth mental health are central themes. The model predicts entailment between each article and the hypothesis: *“This article discusses video-based social media platforms (YouTube, TikTok, Snapchat, Instagram Reels, or YouTube Shorts) AND their relationship to mental health among children, adolescents, teenagers, or youth.”*

Articles scoring above 0.5 are retained. A keyword fallback mechanism rescues borderline articles (0.40-0.50) if they contain terms from all three categories: video platforms, youth terms, and mental health terms. This process retained 864 articles (99.3%).

3.4 Multi-Label Topic Annotation

We define 33 mental health topics including anxiety, depression, social media, screen time, bullying, access to care, and others. Manual annotation of 864 articles across 33 topics would require 28,512 labeling decisions, taking approximately 237 hours of expert time. To address this, we first use zero-shot classification with BART-MNLI[1] to generate initial annotations automatically. The model computes entailment scores for each article-topic pair using hypothesis templates like *“This article is about anxiety, worry, panic, stress disorder.”* Topics with scores above 0.5 are assigned to the article.

After automated annotation, we manually verified a subset of the labels to ensure quality and adjusted the threshold where necessary. This hybrid approach produced an average of 10.56 topics per article, with only 3 articles (0.3%) receiving no labels.

3.5 Classification Models

3.5.1 Baseline: TF-IDF with Logistic Regression

We use TF-IDF vectorization with 5,000 maximum features, (1,2) n-grams, and frequency thresholds (min DF=2, max DF=0.8). One-vs-Rest Logistic Regression with balanced class weighting trains 33 independent binary classifiers. We split data 80/20 for training and testing. This baseline is computationally efficient and handles class imbalance through automatic weight adjustment.

3.5.2 Standard Transformer: DistilBERT

We use `distilbert-base-uncased` with a 33-neuron classification head. The data is split 60/20/20 for training, validation, and testing. Training uses learning rate 2×10^{-5} , batch size 8, max length 256 tokens, 5 epochs, dropout 0.1, and standard Binary Cross-Entropy loss with 0.5 threshold for all topics. Early stopping monitors validation macro F1 with patience of 3 epochs.

3.5.3 Improved Transformer

To address class imbalance, we implement two improvements: (1) Weighted Binary Cross-Entropy loss using inverse frequency weighting $w_j = \min(10.0, \max(0.5, \frac{N}{2(n_j+1)}))$, and (2) per-topic threshold optimization on the validation set, testing thresholds from 0.05 to 0.95 in 0.05 steps.

We test two configurations (DistilBERT with LR 1×10^{-5} and BERT-base with LR 8×10^{-6}), selecting the best based on validation macro F1. We use a 70/15/15 split to provide more training data for rare topics.

3.6 Challenges and Solutions

Annotation Cost: Manual labeling was impractical. We used zero-shot learning with BART-MNLI to automate annotation, then manually verified labels for quality assurance.

Class Imbalance: The standard transformer predicted only 8 of 33 topics. We implemented weighted loss and threshold optimization, increasing coverage to 31 topics.

Limited GPU Memory: We used DistilBERT[2] instead of BERT, employed gradient accumulation, and used mixed precision training.

4 Datasets and Experiments

4.1 Dataset Description

4.1.1 Data Sources and Collection

Our dataset consists of news articles collected from two primary sources focusing on video-based social media platforms: YouTube-related news (1,326 articles) and Snapchat-related news (676 articles). These articles were collected in JSONL format and span discussions about video platforms (YouTube, TikTok, Snapchat, Instagram Reels, YouTube Shorts) and their relationship to youth mental health.

4.1.2 Data Statistics

Table 1 summarizes the dataset statistics at each processing stage.

Table 1: Dataset statistics at each processing stage		
Stage	Articles	Retention Rate
Raw collection	1,998	100%
After deduplication	1,702	85.2%
After manual cleaning	870	43.5%
After relevance filtering	864	99.3% (of cleaned)

The final dataset of 864 articles contains an average of 10.56 topics per article. The distribution of topics per article ranges from 0 (3 articles) to 28 topics, with most articles having between 7-14 topics. This high label density reflects the multi-faceted nature of mental health discourse.

4.1.3 Topic Distribution

Table 2 shows the frequency of the top 10 and bottom 5 topics in our dataset.

The extreme imbalance is evident: the most frequent topic (`social_media`) appears 87 times more often than the least frequent (`foster_care`). This nearly two-order-of-magnitude range poses a significant challenge for classification models.

4.1.4 Data Splits

We use different data splits for different models based on their requirements:

- **TF-IDF Baseline:** 80% training (691 articles), 20% testing (173 articles)

Table 2: Topic frequency distribution (top 10 and bottom 5)

Topic	Articles	Percentage
<i>Most Frequent Topics</i>		
social_media	786	91.0%
resilience/coping	731	84.6%
screen_time	709	82.1%
minority_youth	701	81.1%
access_to_care	671	77.7%
peer_support	600	69.4%
family/peers	595	68.9%
trauma/ptsd	532	61.6%
crisis_intervention	400	46.3%
early_intervention	352	40.7%
<i>Least Frequent Topics</i>		
medication	30	3.5%
eating_disorders	17	2.0%
sleep_issues	14	1.6%
foster_care	9	1.0%

- **Standard Transformer:** 60% training (518 articles), 20% validation (173 articles), 20% testing (173 articles)
- **Improved Transformer:** 70% training (605 articles), 15% validation (130 articles), 15% testing (129 articles)

The improved transformer uses more training data to provide better coverage of rare topics. All splits maintain similar topic distributions through stratified sampling where possible.

4.2 Experimental Setup

4.2.1 Evaluation Metrics

We evaluate models using four metrics described in Section 2:

Macro F1 is our primary metric because it treats all topics equally, revealing performance on minority classes. Micro F1 tends to be dominated by frequent topics.

4.2.2 Model Configurations

TF-IDF Baseline: Vocabulary size 5,000; (1,2)-grams; $C = 1.0$; balanced class weighting.
Standard Transformer: distilbert-base-uncased; LR 2×10^{-5} ; batch size 8; max length 256; 5 epochs (early stopping); dropout 0.1; unweighted Binary Cross-Entropy; fixed 0.5 threshold.

Improved Transformer: distilbert-base-uncased; LR 1×10^{-5} ; batch size 8; max length 256; 6 epochs (early stopping); dropout 0.05; **Weighted** Binary Cross-Entropy; **optimized** per-topic thresholds (0.05–0.95).

4.3 Results and Analysis

4.3.1 Overall Performance Comparison

Table 3 compares the three models across all evaluation metrics.

Table 3: Overall performance comparison across models

Metric	TF-IDF	Standard Trans.	Improved Trans.
Exact Match	4.05%	0.58%	0.00%
Hamming Loss	0.1620	–	0.2893
Micro F1	0.7529	0.6565	0.6575
Macro F1	0.5913	0.2094	0.5071

Key Findings:

- The TF-IDF baseline achieves the highest macro F1 (0.5913) and predicts 30 of 33 topics successfully, demonstrating that classical methods remain competitive for this task.
- The standard transformer severely underperforms (macro F1: 0.2094), predicting only 8 of 33 topics. This dramatic failure is due to class imbalance: the model optimizes for frequent topics and ignores rare ones.
- The improved transformer recovers most performance (macro F1: 0.5071), successfully predicting 31 of 33 topics. This represents a 142% improvement in macro F1 and 287.5% improvement in topic coverage compared to the standard transformer.
- All models struggle with exact match, reflecting the difficulty of predicting all 10+ labels correctly for each article.
- Micro F1 scores are similar across models (0.65–0.75), indicating that all models perform reasonably well on frequent topics.

4.3.2 Per-Topic Performance Analysis

Figure 1 shows F1-scores for all topics across the 2 best models. Table 4 presents detailed results for selected topics.

Analysis:

- For frequent topics (>100 samples), all three models achieve $F1 > 0.85$, with similar performance.

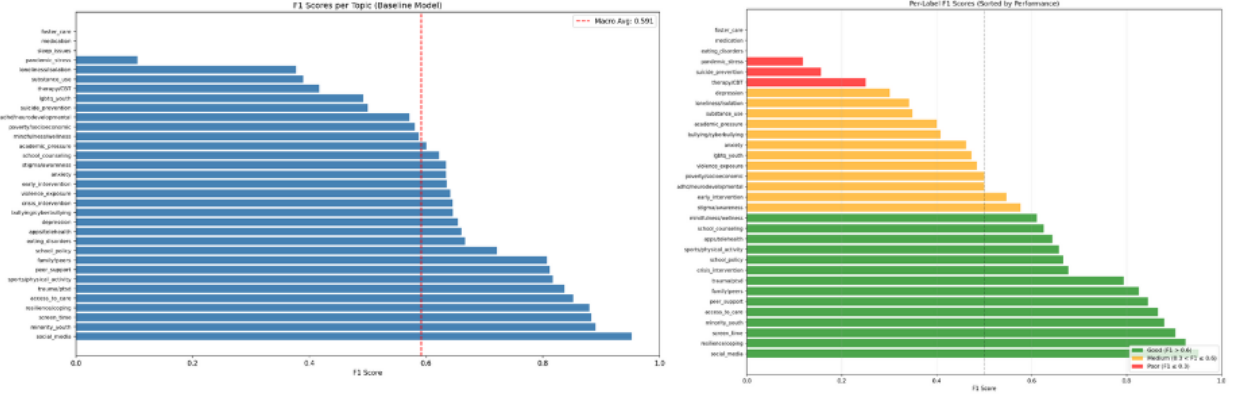


Figure 1: Comparison of F1 scores across different topics.

Table 4: Per-topic F1-scores for selected topics

Topic	Support	TF-IDF	Standard	Improved
social_media	156/118	0.952	0.948	0.952
resilience/coping	143/110	0.880	0.905	0.924
screen_time	143/107	0.883	0.905	0.903
minority_youth	138/102	0.890	0.887	0.879
access_to_care	130/98	0.852	0.858	0.866
anxiety	38/28	0.634	0.000	0.462
depression	29/23	0.655	0.000	0.301
substance_use	36/26	0.390	0.000	0.348
eating_disorders	3/2	0.667	0.000	0.000
medication	3/2	0.000	0.000	0.000
foster_care	1/1	0.000	0.000	0.000

- For medium-frequency topics (20-100 samples), the TF-IDF baseline performs best, while the improved transformer shows moderate success and the standard transformer fails completely.
- For rare topics (< 20 samples), all models struggle. The improved transformer achieves non-zero F1 for more topics but with lower scores than the baseline.
- Topics with fewer than 3 test samples (eating_disorders, medication, foster_care) are unpredictable by all models, likely due to insufficient training examples.

4.3.3 Hyperparameter Sensitivity

We tested two transformer configurations for the improved model:

The DistilBERT configuration with slightly higher learning rate performed best. BERT-base did not provide sufficient improvement to justify its increased computational cost.

Table 5: Hyperparameter search results

Configuration	Model	Learning Rate	Val Macro F1
Config 1	DistilBERT	1×10^{-5}	0.3909
Config 2	BERT-base	8×10^{-6}	0.3654

4.3.4 Topic Co-occurrence Analysis

Table 6 shows the most frequent topic pairs.

Table 6: Top 5 topic co-occurrences

Topic Pair	Co-occurrence Count
resilience/coping + social_media	688
screen_time + social_media	676
minority_youth + social_media	676
access_to_care + social_media	640
minority_youth + resilience/coping	632

Nearly all articles discussing social media also address coping mechanisms (79.6%) and screen time effects (78.2%), reflecting the interconnected nature of mental health discourse.

4.3.5 Error Analysis

Manual inspection of misclassified examples reveals common error patterns:

False Negatives (missed labels): Articles discussing topics implicitly or using indirect language (e.g., referring to "wellness programs" instead of explicitly mentioning "therapy") are often missed, especially for rare topics.

False Positives (over-prediction): Generic mental health articles mentioning social media in passing are sometimes assigned all major topics, even if only one or two are truly central.

Boundary Cases: Articles discussing adjacent topics (e.g., "body image" vs. "eating disorders") create ambiguity, particularly when the zero-shot model’s score falls near the threshold.

4.4 Visualization Analysis

We apply dimensionality reduction (UMAP) to transformer embeddings to visualize article clusters. Figure 2 shows the 2D projection of all 864 articles, colored by primary topic.

Key observations from clustering:

- Articles with similar topic combinations cluster together, validating that the model learns meaningful representations.
- The largest cluster corresponds to articles discussing social media, screen time, and coping/resilience (the three most co-occurring topics).



Figure 2: An interactive HTML [3] version of this plot is available in the attached references

- Rare topics like foster care and eating disorders do not form distinct clusters, explaining why models struggle to identify them.
- Clear separation exists between intervention-focused articles (therapy, counseling) and symptom-focused articles (anxiety, depression), despite both being mental health topics.

4.5 Discussion

Our experiments demonstrate several important findings:

Classical methods remain competitive: The TF-IDF baseline outperforms both transformer models on macro F1, likely due to the relatively small dataset size (864 articles) where feature engineering and explicit handling of class imbalance provide advantages over end-to-end learning.

Class imbalance requires explicit handling: The dramatic failure of the standard transformer (predicting only 8 of 33 topics) highlights that modern neural architectures do not automatically handle imbalanced data. Weighted loss and threshold optimization are essential.

Rare topics remain challenging: Despite improvements, all models achieve low F1-scores on topics with fewer than 20 test samples. This suggests that either (1) more data is needed, or (2) these topics require different modeling approaches such as few-shot learning.

Zero-shot annotation is effective but imperfect: The BART-MNLI annotation achieved reasonable quality, enabling this project without manual labeling. However, some annotation errors likely propagate through to model training, potentially limiting performance.

Multi-label complexity: The average of 10.56 topics per article and high topic co-occurrence make this a genuinely difficult multi-label problem. Exact match rates near zero across all models reflect this inherent difficulty.

5 Conclusions

5.1 Limitations

Several limitations should be acknowledged:

Dataset Scope: Our dataset focuses on only two source categories (YouTube and Snapchat news) and may not generalize to broader social media and mental health discourse. The 43.5% retention rate after manual cleaning indicates potential sampling bias.

Zero-Shot Annotation Quality: While practical, automated annotation using BART-MNLI may contain errors that propagate to model training. Comprehensive human validation was performed.

Rare Topic Performance: Topics with fewer than 20 samples (eating disorders, medication, foster care, sleep issues) achieve near-zero F1-scores across all models. These topics require either more data or alternative approaches like few-shot learning.

Exact Match Difficulty: All models achieve near-zero exact match rates, indicating that predicting all 10+ labels correctly per article remains extremely challenging. This may limit practical applicability for tasks requiring perfect label sets.

5.2 Future Work

Several directions could improve upon this work:

Data Augmentation: Expanding the dataset to include more sources (TikTok, Instagram Reels news) and conducting comprehensive human annotation for rare topics would improve model performance and reliability.

Few-Shot Learning: For extremely rare topics, few-shot learning approaches like prototypical networks or meta-learning could be more effective than standard supervised learning.

Hierarchical Classification: Organizing the 33 topics into a taxonomy (e.g., symptoms, interventions, social factors) and using hierarchical multi-label classification could leverage relationships between topics.

5.3 Conclusion

This project successfully demonstrated a pipeline for multi-label topic classification in a specialized domain with limited resources. While modern transformer models show promise, they require careful handling of class imbalance to avoid severe underfitting. Classical methods remain competitive and should not be overlooked, especially for smaller datasets. The insights gained about news media coverage patterns, particularly the underrepresentation of critical topics like foster care and eating disorders, highlight areas where public health communication could be strengthened. Overall, this work provides a foundation for automated analysis of mental health discourse in news media and demonstrates practical techniques for addressing common challenges in multi-label classification tasks.

6 Project Management

6.1 Roles and Responsibilities

6.1.1 Author 1: Ali Abdullah Ahmad

Primary Responsibilities:

Data Preprocessing: Implemented the full preprocessing pipeline, including raw data loading, text normalization, MD5-based deduplication, and cleaning of malformed or incomplete records.

Binary Relevance Classification: Implemented the BART-MNLI zero-shot classifier for determining whether each article was related to video-Based social media AND youth mental health.

Baseline Model Development: Developed the TF-IDF with Logistic Regression baseline model, configured One-vs-Rest classification with class weighting, and produced evaluation metrics and diagnostic plots.

Transformer Fine-Tuning: Implemented and trained the fine-tuned transformer classifier, including class imbalance handling, weighted BCE loss, and per-topic decision threshold optimization.

6.1.2 Author 2: Rishab Apkari

Primary Responsibilities:

Manual Filtering: Performed manual review and refinement of the dataset to validate article relevance, remove borderline content, and ensure quality of the final corpus.

Multi-Label Annotation Pipeline: Implemented the multi-label topic annotation system using BART-MNLI. Designed hypothesis templates for all 33 topics, calculated the entailment probabilities for each pair of articles - topic pair, and applied a calibrated thresholding mechanism to assign multiple labels. Coordinated with Author 1 to manually verify a subset of annotations.

Transformer Model Development: Implemented the standard transformer pipeline (non-finetuned version), including dataset preparation, training routines, and comparison experiments across multiple architectures.

Visualization and Analysis: Developed the visualization pipeline (UMAP + transformer embeddings) and produced both interactive and static plots. Conducted error analysis, topic co-occurrence analysis, and identified failure modes.

6.2 Collaborative Efforts

Both authors jointly contributed to the following areas:

Multi-Label Annotation: performed manual verification of assigned labels.

Topic Selection: Defined the set of 33 mental health topics using BARTopic clustering outputs combined with manual refinement.

Experimental Design: Designed the experimental protocol, including data splitting strategy, evaluation metrics, and model comparison framework.

Analysis of Results: Jointly interpreted model performance, identified failure cases, and proposed solutions related to class imbalance and thresholding.

Code Reviews: Reviewed each other’s code to ensure reproducibility, clarity, and correctness across the entire project pipeline.

Final Report Preparation: Collaboratively edited, cross-checked, and finalized the report, ensuring consistency and correctness of all results.

References

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